Shivansh Rajput

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github.com/RAPTOR-sr | Portfolio Website

Summary

AI/ML-focused computer science student with practical experience in computer vision, gesture recognition, and LLM-based applications. Adept in Python, TensorFlow, PyTorch, and natural language processing, with an emphasis on developing scalable AI solutions and high-accuracy machine learning models.

Education

Manipal University Jaipur, Jaipur — Sep 2022 – Present
Bachelor of Technology (BTech) - Computer Engineering

- GPA: 7.0 / 10.0

Birla Public School, Pilani — Jan 2021 – March 2022
High School (PCM+IT Stream)

- Percentage: 89%

Skills

Programming Languages: Python, C++, SQL
AI/ML: Machine Learning, Deep Learning, NLP, LLMs, RAG, Computer Vision, CNN, Object Detection (YOLO)
Cloud & DevOps: Oracle Cloud Infrastructure (OCI), Git, GitHub, GitHub Actions (CI/CD), Docker, Nginx
Web Technologies: Streamlit, FastAPI, Django, HTML5, CSS3
Operating Systems & Tools: Linux (Ubuntu), VS Code, SSH, Cron Jobs, Bash
Databases & Data: MySQL, Pandas, NumPy, Pydantic
Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-learn, OpenCV, LangChain, FAISS, Matplotlib, Plotly

Experience

Python & Cloud Developer Intern — Feb 2026 – Jul 2026
BabaBuilds

- Developed and deployed AI-powered web applications while managing cloud infrastructure, Linux servers, and secure production environments.
- Configured Oracle Cloud Infrastructure, Nginx, SSL/TLS, GitHub Actions CI/CD, and server security using CrowdSec and Linux tools.

AI/ML Intern — May 2025 – July 2025
CSIR - CEERI, Pilani

- Captured and processed high-precision 3D point clouds for 59 static and dynamic hand gestures using Time-of-Flight (ToF) camera technology.
- Engineered a real-time hand detection, segmentation, and visualization pipeline, achieving 25–30 FPS on an RTX 3060 (12GB) GPU with mAP@0.5:0.95 - 0.85, Precision: 98%, and Recall: 97%.
- Led a 4-member team over 45 days to collect and structure a 59-class gesture dataset containing 22,000+ RGB images and point cloud samples, supporting future model training and evaluation..

Projects

AnalyticsHub – AI-Powered Document Intelligence Platform — Live Demo

- Developed an AI-powered document intelligence platform using Python, Streamlit, and Groq LLMs.
- Deployed and secured the application on Oracle Cloud Infrastructure using Ubuntu, Nginx, SSL/TLS, and CrowdSec.
- Built document summarization, factual validation, entity extraction, tone analysis, and interactive AI document chat.
- **Technologies Used:** Python, Streamlit, Groq API, Llama 3.3, Oracle Cloud (OCI), Ubuntu Linux, Nginx, GitHub Actions, CrowdSec

Real-Time CCTV Object Detection & Logging — GitHub Link

- Developed a Python-based live video analysis system, achieving 20–30 FPS on a laptop GPU to detect and label up to

80 object categories (e.g., cars, trucks, people) using YOLOv8.
- Achieved mAP@0.5 - 0.92, Precision: 96%, and Recall: 95%, enabling high-accuracy detection and reducing manual CCTV review time by over 70
- Integrated BLIP to generate contextual captions for detected objects and log results with timestamps to a CSV file for fast event retrieval.
- **Technologies Used:** YOLO v8, BLIP, OpenCV, Tkinter, PyTorch, Transformers.

Sign-Language-to-Text Conversion GitHub Link
- Built a machine learning model to translate 28 sign language gestures into text, trained on 8,400 webcam-captured images, achieving 70% accuracy.
- Implemented real-time gesture recognition using computer vision techniques, delivering 30–40 FPS for smooth live translation.
- **Technologies Used:** Python, Machine Learning, Computer Vision, OpenCV, Webcam Data Processing

CNN-based Image Classification Model for Skin Disease Identification
- Implement a Convolutional Neural Network (CNN) model using TensorFlow and Keras to identify potential skin diseases from images.
- Trained the model on a custom dataset, achieving 75% accuracy with ongoing optimization to enhance performance.
- **Technologies Used:** Python, CNN, TensorFlow, Keras, Image Classification, Custom Dataset Handling

Certifications

Design and Analysis of Algorithms – NPTEL Oct 2024
Tableau Fundamentals – Salesforce Nov 2024
Switching, Routing, and Wireless Essentials – Cisco Networking Nov 2024
Data Structures – Coursera Dec 11, 2023
Programming in Python – Meta (via Coursera) Nov 21, 2023

Interests

Artificial Intelligence, Machine Learning, Computer Gaming, Badminton, Traveling

Languages

English (Fluent), **Hindi** (Native)